4.Aim:

To solve the Crypt-Arithmetic problem using Python.

Algorithm:

1. Assign different digits to different letters.
2. Check whether the given arithmetic equation is correct.
3. Display the valid digit assignment.
4. Stop.

Code:

```
from intercools import permutations
letters = "SENDMORY"
for p in permutations(range(10), len(letters)):
    d = dict (zip (letters, p))
    if d["S"] == 0 or d["M"] == 0:
        continue
    send = d["S"]*1000 + d["E"]*100 + d["N"]*10 + d["D"]
    more = d["M"]*1000 + d["O"]*100 + d["R"]*10 + d["E"]
    money = (d["M"]*10000 + d["O"]*1000 +
            d["N"]*100 + d["E"]*10 + d["Y"])

    if send + more == money:
        print ("SEND =", send)
        print ("MORE =", more)
        print ("MONEY =", money)
        break
```

Output:

```
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SEND = 9567
MORE = 1085
MONEY = 10652
```

Result:

Thus, the Crypt-Arithmetic problem was solved successfully using Python.